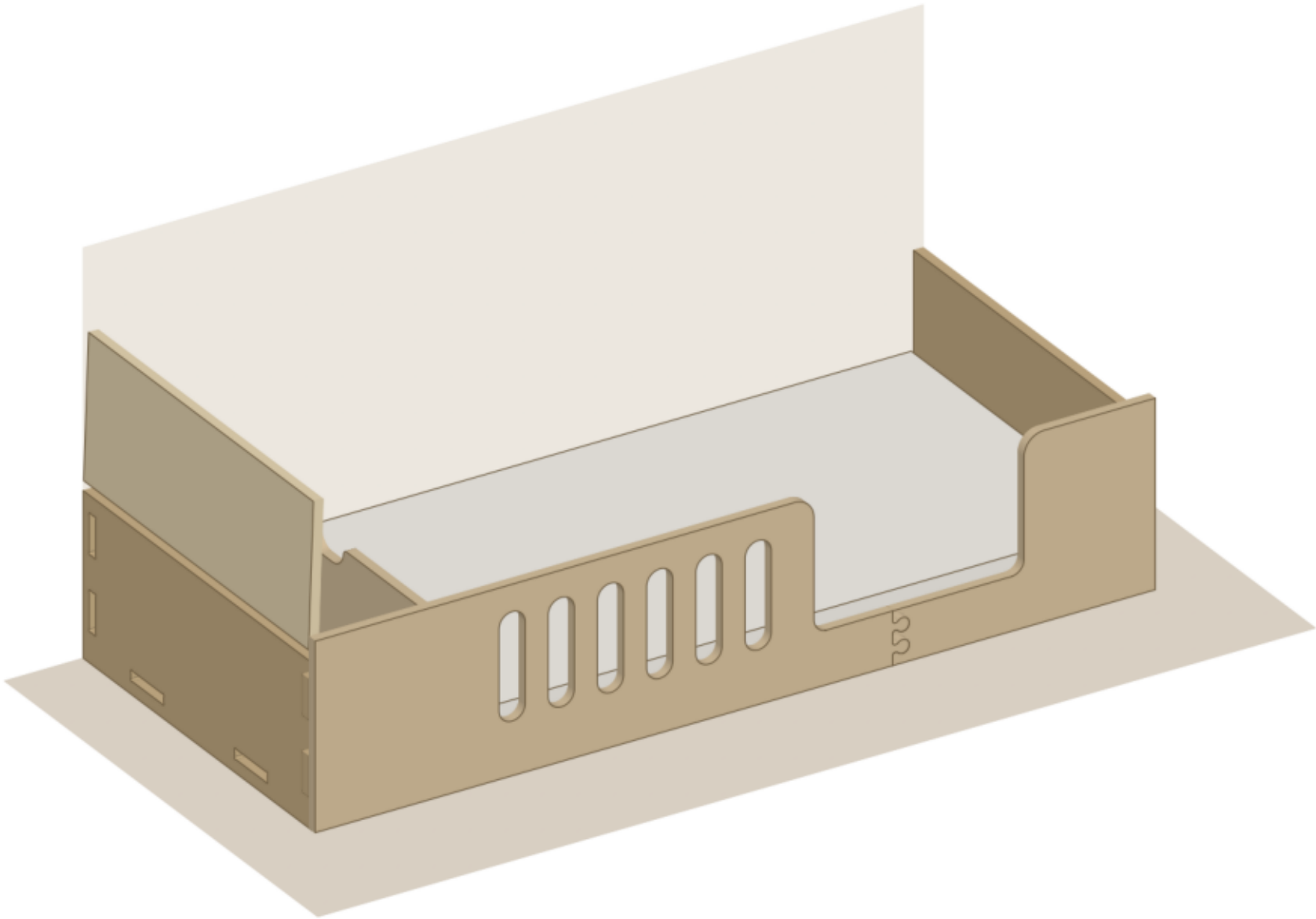
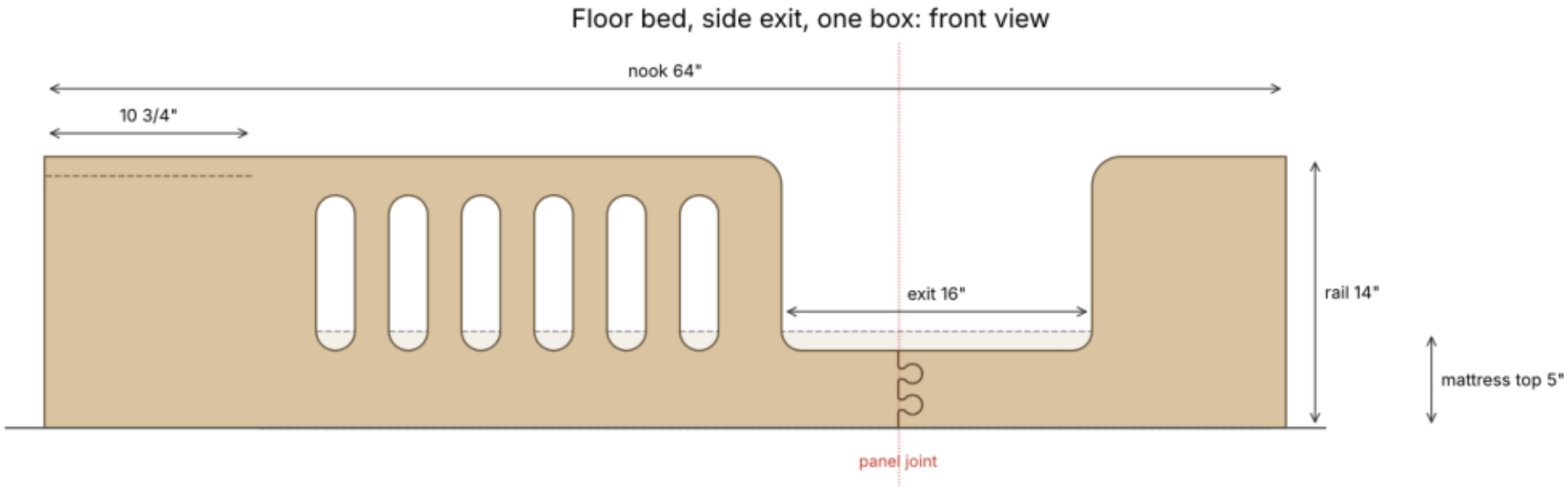
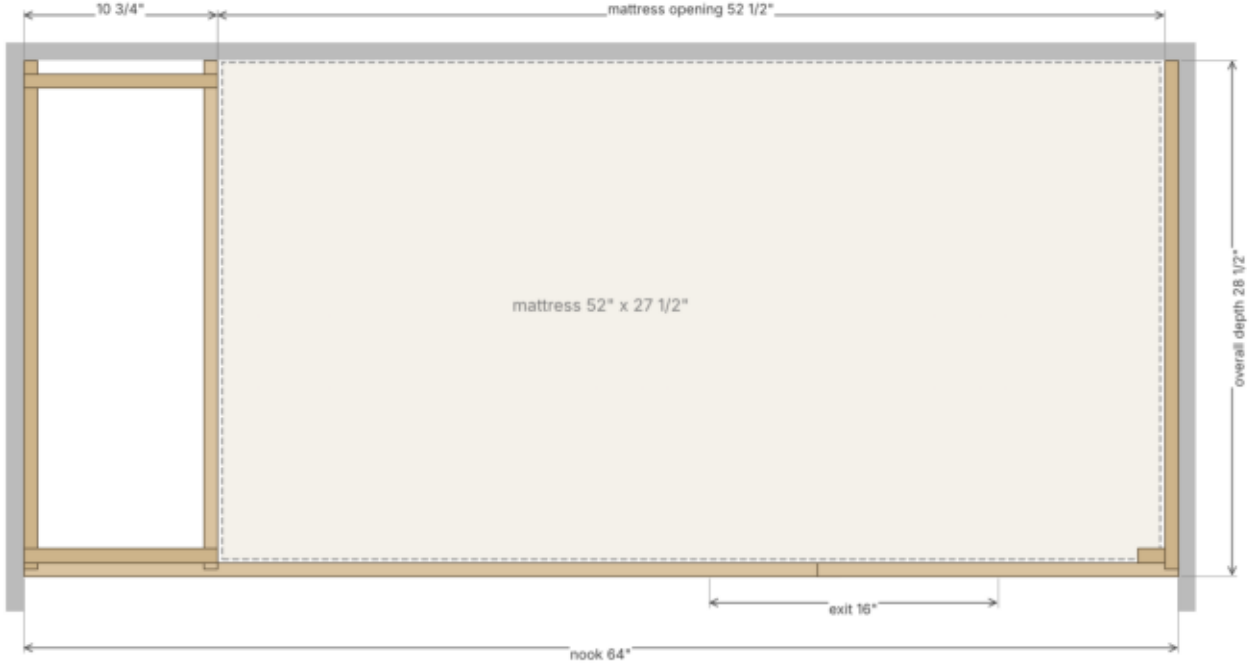
Floor bed, side exit, one box

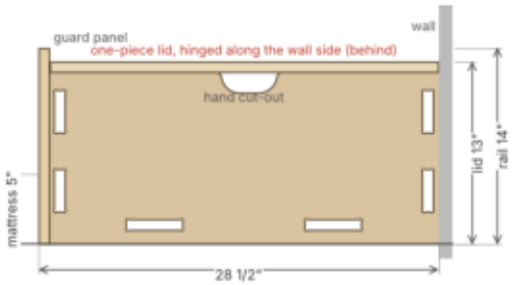



Floor bed, side exit, one box: front view
nook 64"
10 3/4"
exit 16"
rail 14"
mattress top 5"
panel joint

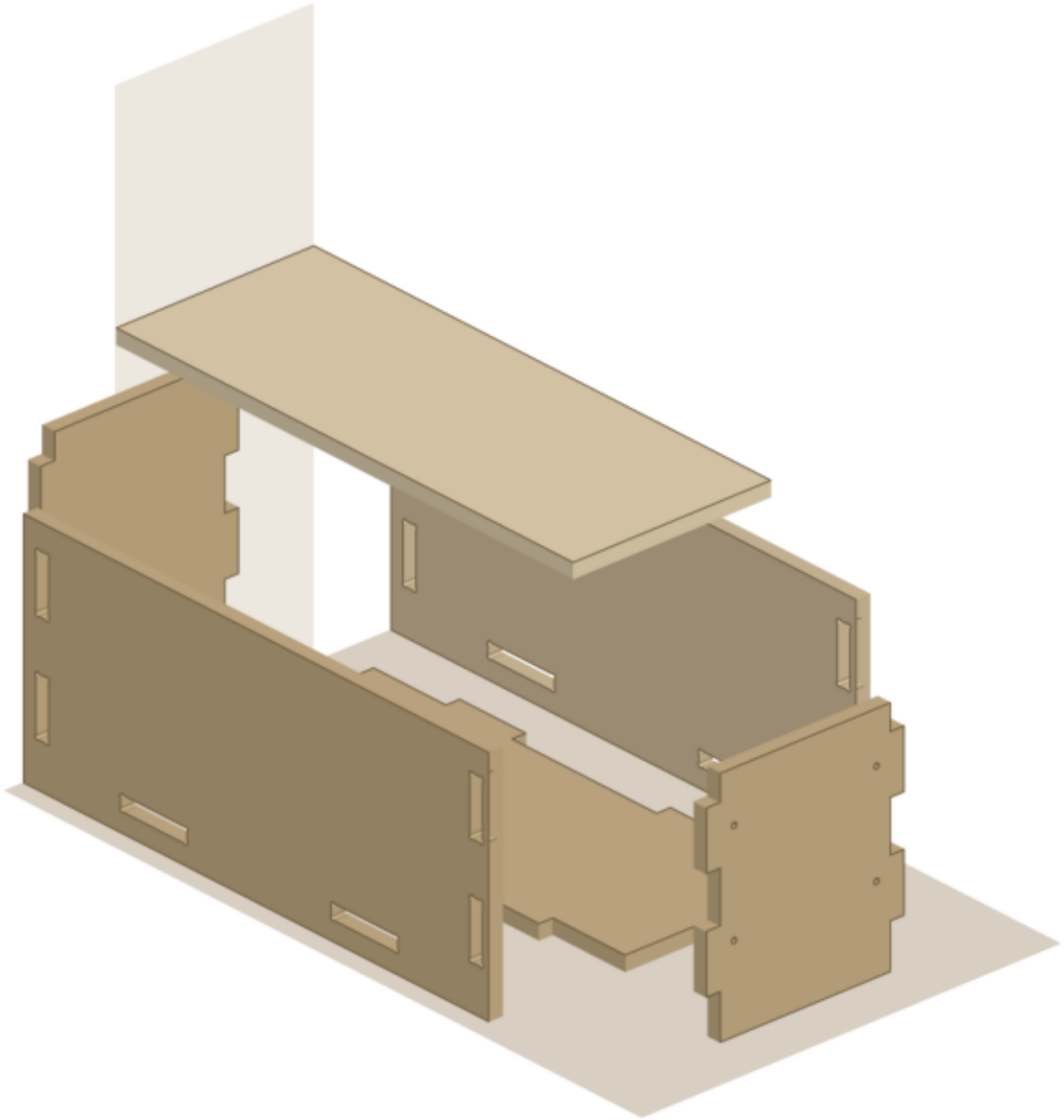
Plan (front of bed at the bottom)



Head box seen from the mattress

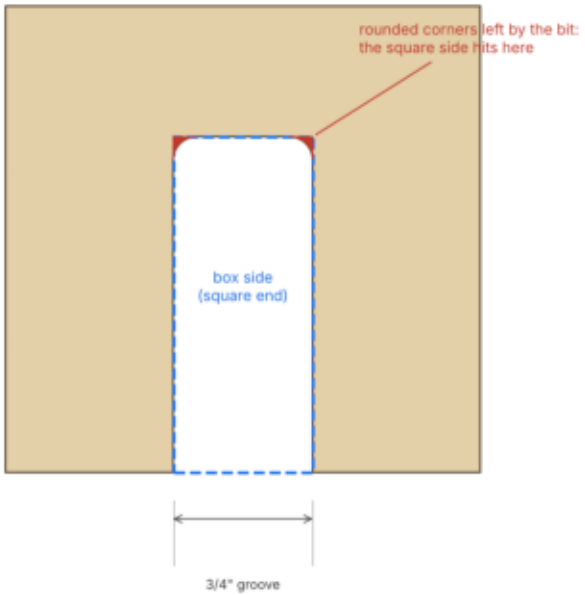


Head box, exploded (tabs pass through the sides)

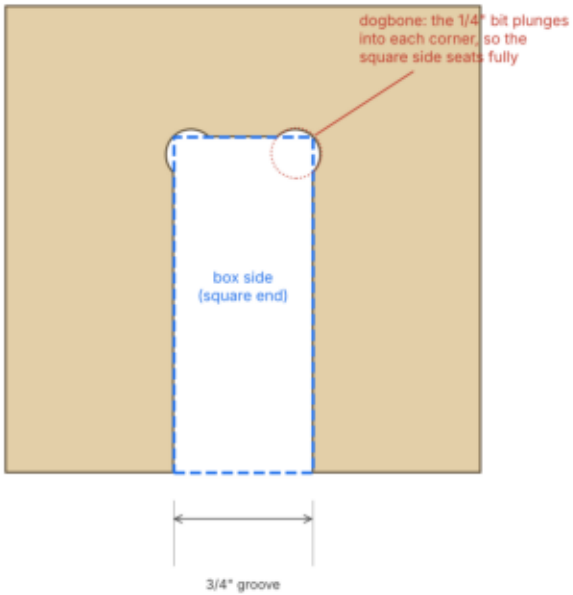


Dogbones: how a square part fits a CNC-cut groove (looking at the back of the front panel)

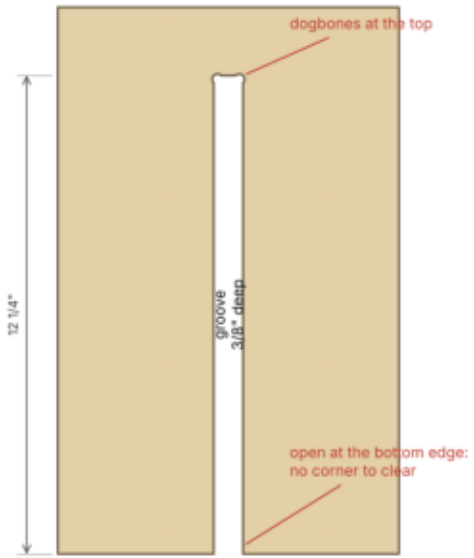
Without dogbones (red = wood that blocks the square corner)



With dogbones

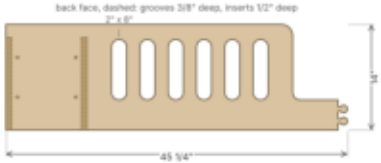


Back of the front panel, one groove

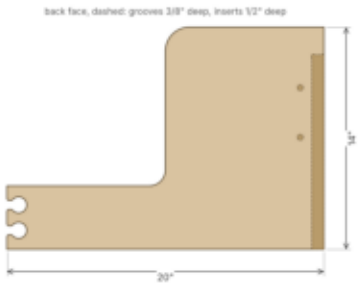


Floor bed, side exit, one box: part drawings (inches, 3/4" plywood)

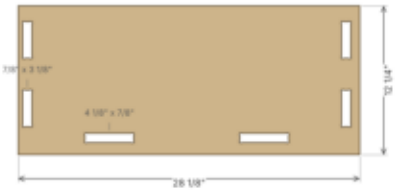
Front panel A (qty 1)
knob half



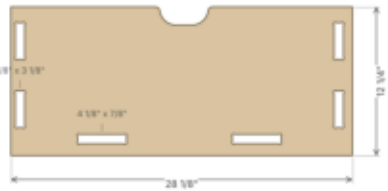
Front panel B (qty 1)
socket half



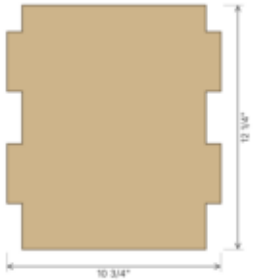
Box side (wall) (qty 1)
against the nook wall, 1 per box



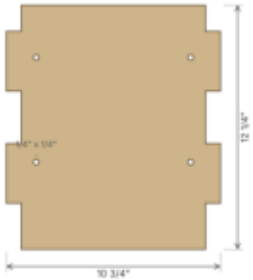
Box side (mattress) (qty 1)
faces the mattress; hand cut-out to open the lid



Box back (qty 1)
screw through into wall studs



Box front (qty 1)
against the back of the front panel; 4 bolts into it



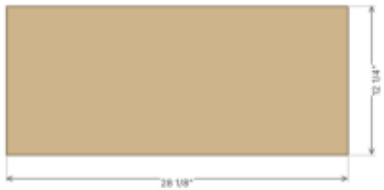
Box floor (qty 1)
sits 1" off the floor



Box lid (qty 1)
piano hinge along the long wall-side edge



Foot wall liner (qty 1)
screw to the foot wall studs; runs into the panel groove

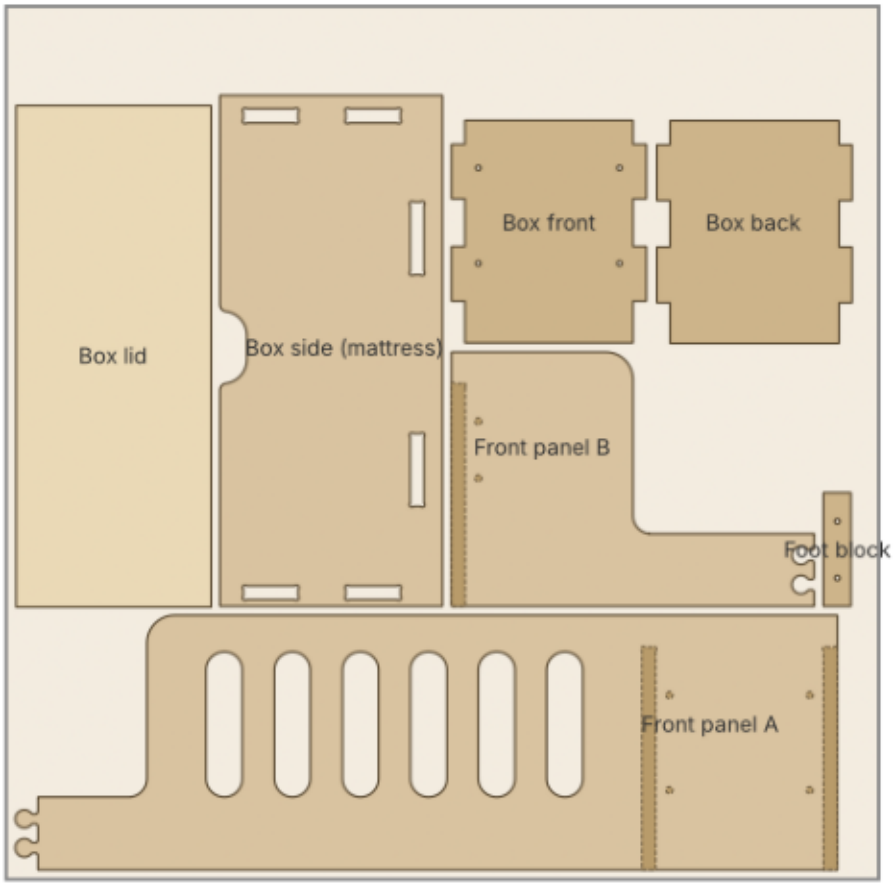


Foot block (qty 1)
above the mattress; glue + 2 screws to the liner, 2 bolts into the panel



Floor bed, side exit, one box: nesting on 48" square panels

4'x4' panel 1



4'x4' panel 2



Floor bed, side exit, one box: cut list, hardware, assembly

Cut list

Part	Qty	Size (W x H)	Notes
Front panel A	1	45 1/4" x 14"	knob half
Front panel B	1	20" x 14"	socket half
Box side (wall)	1	28 1/8" x 12 1/4"	against the nook wall, 1 per box
Box side (mattress)	1	28 1/8" x 12 1/4"	faces the mattress; hand cut-out to open the lid
Box back	1	10 3/4" x 12 1/4"	screw through into wall studs
Box front	1	10 3/4" x 12 1/4"	against the back of the front panel; 4 bolts into it
Box floor	1	10 3/4" x 25 1/2"	sits 1" off the floor
Box lid	1	10 3/4" x 27 5/8"	piano hinge along the long wall-side edge
Foot wall liner	1	28 1/8" x 12 1/4"	screw to the foot wall studs; runs into the panel groove
Foot block	1	1 1/2" x 6 1/4"	above the mattress; glue + 2 screws to the liner, 2 bolts into the panel

Key dimensions

Panels: 2 pieces of 4'x4' (1 x 4'x8' sheet(s) of 3/4" plywood, each cut in half)
Heights from floor: mattress top 5", rail top 14", box lids 13", exit sill 4"
Exit opening: 38" to 54" from the head wall; panel halves join at 44"
Head box: 10 3/4" wide outside, 9 1/4" inside, 27 3/4" deep; no foot box (mattress against a liner on the foot wall)

Hardware

- Wood glue (PVA)
- 1-1/4" wood screws x 14 (box joinery + foot block)
- 1/4-20 threaded inserts for wood x 6, no longer than 3/8" (front panel)
- 1/4-20 x 1" bolts + washers x 6 (box fronts and foot block to front panel)
- 2-1/2" screws x 5 into studs (box back and foot liner)
- Piano hinge 1-1/2" open width, 1 x 24" length(s) (1 per lid)
- Soft-close lid stays x 1 (light-duty toy-box type)
- Shims as needed

Assembly

1. Cut all parts (sheets/panel_*.dxf). The two front-panel pieces are mirrored in the files: cut them BACK FACE UP so the POCKET layers (box-side grooves 3/8" deep, insert holes 1/2" deep) land on the back. Profile CUT_INSIDE before CUT_OUTSIDE, use tabs/onion skin on small parts, and dry-fit the puzzle joint and one groove on scrap first; adjust knob_clearance / tab_clearance if needed.
2. Ease every exposed edge with a 1/4" roundover (rail top, slots, exit opening, lid edges and the